

Technical Document #723: Blockchain - Comprehensive Overview

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Cross-chain technology enables communication between different blockchains. It is essential for interoperability in the blockchain ecosystem.

Smart contracts are self-executing contracts with terms directly written into code. They automatically enforce agreements without intermediaries.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Cross-chain technology enables communication between different blockchains.
- Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain.
- Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions.